

AD7190

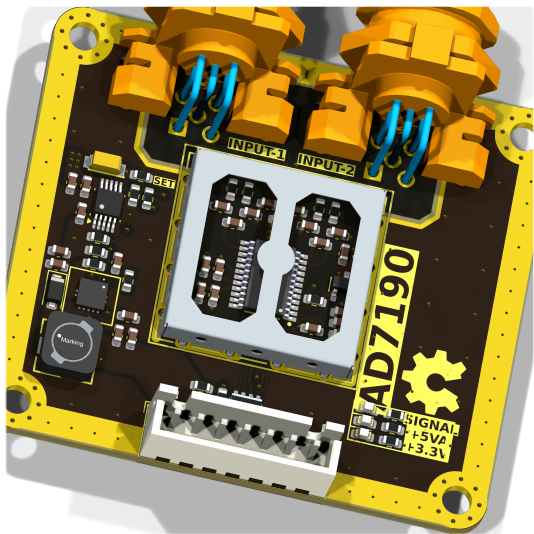
Variant:

2026-09-29

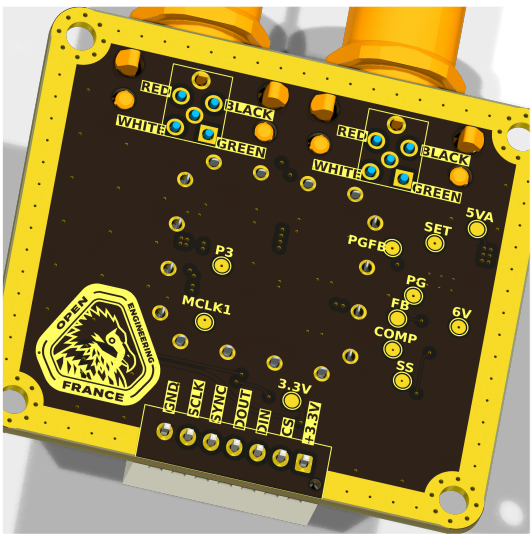
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

OPEN_WEIGHT - AD7190 24-bit load-cell front-end, part of OPEN THRUST LAB.
Single +3.3 V host supply, SPI interface, 2 x 4-wire bridge inputs.

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 29-Sep-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:
+3.3V host ->
TPS61086
6 V -> LT3042 +5VA
(AVDD + excitation)

DEBUG NOTE:
TP on every rail.
Wait >= 600 ms (PG)
before ADC init.

DESIGN NOTE:
SPI mode 3,
SCLK <= 5 MHz,
RDY on DOUT.

DESIGN NOTE:
REFIN1 = bridge
excitation:
ratiometric.

LAYOUT NOTE:
AFE under the
WE-SHC shield H5,
boost SW node
away.

	Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
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